Uncertainty Signal Pilot: Perception and Reasoning Entropy

FERMAT Handwritten Math, Qwen2.5-VL-3B

August 5, 2026

Abstract

This study tests whether two entropy-based uncertainty signals – *perception entropy* (instability in transcribing a handwritten math answer) and *reasoning entropy* (instability in grading whether an answer contains an error) – predict model correctness, using Qwen2.5-VL-3B on the FERMAT dataset. It ran in three phases: an exploratory $n = 100$ pilot, a confirmatory $n = 300$ run with a pre-registered analysis and a deliberately rebalanced label distribution, and a follow-up closing the two gaps that run left open. The two arms end in opposite places, and both results are firm.

Perception entropy works. At $n = 300$ it reaches AUROC $= 0.835$, 95% CI $[0.787, 0.879]$, replicating the pilot's 0.788 and tightening the interval by half. It survives every artifact check: 0.796 $[0.736, 0.852]$ after removing all parse failures *and* all maximum-entropy items simultaneously. A simple decision rule falls out of it – when all five transcriptions disagree, the model is wrong 57 times out of 61 (93.4% precision). It also beats an independent baseline: the model's own token log-probabilities reach only AUROC 0.537, a resolved paired gap of +0.297 $[+0.214, +0.380]$ (§6.1), so the signal is not simply recoverable from information the model already exposes. A further five samples per item (§6.1) push $K = 10$ to AUROC 0.806 against $K = 5$'s 0.761 on a fair, like-for-like label, a resolved +0.045 $[+0.005, +0.087]$.

Reasoning entropy fails, for a reason that is not about entropy. The pilot's apparent 0.618 was an artifact of FERMAT's 87/13 class imbalance: a constant "there is an error" predictor scores 0.87 there, and the model's 75% grading accuracy was losing to it. Rebalancing to 50/50 at $n = 300$ removes that flattery and the model's accuracy collapses to **51.7% against a 0.500 baseline**. It answers "there is an error" on 93% of items – 94.7% correct on error-containing answers, 8.7% on clean ones. Reasoning entropy correspondingly measures nothing: AUROC $= 0.522$ $[0.462, 0.582]$, with 42% of items pinned at zero entropy. The finding is not that the metric is badly implemented but that *sampling entropy cannot predict correctness for a model performing at chance* – there is no competence for disagreement to track. A pre-registered stratified hypothesis, formed post-hoc on the pilot data, proved untestable: the model's bias left only 8 of 150 error items misgraded, below the registered minimum. Three alternative grading prompts were screened against this bias and none fixed it (§6.3); one of them swung the model's response rate from 94% "error" to 38% while leaving accuracy essentially unchanged, evidence that the ceiling is a capability limit rather than a promptable default.

The methodological lesson is separable from either result. Adding bootstrap confidence intervals late overturned several conclusions the pilot had already formalized into code and prose, and the single most consequential design decision was not the metric or the model but the *label balance of the evaluation set*.

1 Goal

Given a vision-language model transcribing and grading handwritten math answers, does the entropy of its own repeated outputs – how much it disagrees with itself across independent samples

– predict when it is actually wrong?

The work ran in three phases, and the distinction matters for how much weight any individual number carries:

- **Phase 1 ($n = 100$, exploratory).** Whether either signal exists at all. Sections 3–4. Several analyses here were chosen after seeing data, and are labelled as such.
- **Phase 2 ($n = 300$, confirmatory).** Section 5. A rebalanced sample with thresholds registered before the run (`pilot.plotting.SCALEUP_PREREGISTRATION`). This is the phase the headline claims rest on. Includes a risk-coverage and conformal deferral analysis of the $n = 300$ result (§5.6).
- **Phase 3 (same $n = 300$ items, follow-up).** Section 6. Closes two gaps Phase 2 left open: an independent baseline for perception entropy, and a screen of alternative grading prompts for the reasoning arm.

Still out of scope across all three: no human double grading, no full-dataset run, and no larger model – the single most important remaining gap, discussed in §5.3 and the final recommendation.

Every AUROC below is reported with a 95% bootstrap confidence interval (10,000 item-level resamples, seed 0; `pilot.plotting.bootstrap_auroc_ci`). These were added late, after the substantive experiments were complete, and they materially changed what this report can claim – several differences it originally presented as findings turn out to be within sampling error at $n = 100$. Where two conditions are compared, the comparison uses a *paired* bootstrap over the same resampled items (`bootstrap_auroc_difference_ci`) rather than checking whether two marginal intervals overlap, which is a strictly weaker and commonly misread test.

2 Setup

- **Data**: FERMAT (2244 items, 84.8% containing an error). Phase 1 drew 100 items at random (seed 42), inheriting that 87/13 split. Phase 2 drew 300 items *stratified* to 150/150 on `has_error` (`pilot.data.load_fermat_balanced`) – for reasons developed in §4.5, this turned out to be the single most consequential design choice in the study.
- **Model**: Qwen2.5-VL-3B-Instruct.
- **Perception**: the model transcribes a handwritten Question–Answer pair; $K = 5$ samples at temperature 0.7, plus 2 samples at temperature 0 as a determinism sanity anchor.
- **Reasoning**: the model grades whether the handwritten Answer contains an error (binary 0/1); same $K = 5 + 2$ temperature-0 sampling scheme.
- **Entropy**: Shannon entropy in nats over the cluster distribution of the K sampled outputs for one item, $H = -\sum_i p_i \ln p_i$.
- **Correctness labels**: transcription is compared against the ground-truth answer; grading is compared against the ground-truth error label.

3 Phase 1: Perception Entropy – Two Failures, One Fix

3.1 Attempt 1: exact-string clustering (failed)

The transcription prompt asks for the *full* handwritten derivation, not just a final answer. Clustering by exact string match after light normalization treated any paraphrasing of that derivation as a different answer.

Items pinned at maximum entropy ($\ln 5 \approx 1.609$)	83/100

83% of items showed the maximum possible disagreement regardless of whether the model actually agreed with itself on the substance of the answer – the metric carried essentially no information.

3.2 Attempt 2: naive semantic clustering (failed)

Reasoning that exact-match was too strict, the next attempt clustered by semantic similarity instead: bidirectional NLI entailment and, separately, embedding cosine similarity, both applied to the whole derivation.

	Canonical (Attempt 1)	NLI	Embedding
Mean entropy	1.535	0.137	0.185
Items at max entropy	83/100	0/100	0/100

This looked like a fix – entropy dropped sharply – but the correctness rate that came with it (majority-vote transcription matched against ground truth via the same NLI entailment check) was $92/97 = 95\%$, implausibly high next to the model's other performance figures. Inspection showed the cause: with premise and hypothesis sharing nearly identical scaffolding (the same LaTeX structure, the same boilerplate phrasing), the NLI model's well-documented lexical-overlap bias pushed it toward "entailment" regardless of whether a buried number or coefficient actually differed. The fix for Attempt 1's problem introduced a new, opposite failure mode: over-merging.

3.3 Attempt 3: final-answer extraction (worked)

The actual defect in both prior attempts was comparing the *wrong span of text*: the full derivation, when only the final answer matters for a correctness judgment. Attempt 3 extracts just the final answer – the last multiple-choice option statement, a `\boxed{}` result, the last display-math block, or (rarest) the last line – before clustering with the same syntactic/canonicalization approach as Attempt 1.

Metric	Value
AUROC (perception entropy $\rightarrow$ transcription error)	**0.788**
95% CI	$[0.697, 0.869]$
Items at max entropy	14/100
Transcription correct	42/100
Median entropy, correct group ($n = 42$)	0.500
Median entropy, incorrect group ($n = 58$)	1.332
Mean entropy, correct group	0.583
Mean entropy, incorrect group	1.099

This is the strongest result in the pilot: real, visible separation between correct and incorrect items (Figure 2, left panel), and the only arm whose confidence interval excludes chance by a clear margin.

3.4 Sensitivity: is 0.788 an artifact?

An entropy AUROC can be manufactured two ways that have nothing to do with the model being usefully uncertain, and both are live concerns here. First, parse failures receive their own cluster label, so an unparseable sample inflates entropy *and* tends to be scored incorrect – the metric would then be measuring the parser rather than the model. Second, $K = 5$ entropy takes only 7 distinct values, and the 14 items pinned at $\ln 5$ are *all* incorrect; if that single cell carried the ranking, there would be no graded signal underneath it. Each cut was rerun with its own CI (`pilot.plotting.auroc_sensitivity`):

Subset	n	AUROC	95% CI
Full	100	0.788	$[0.697, 0.869]$
Excluding all parse failures	85	0.773	$[0.672, 0.865]$
Excluding max-entropy items	86	0.721	$[0.611, 0.822]$
Excluding both	74	**0.707**	$[0.587, 0.819]$

Neither cut explains the result. Dropping all 15 parse-failure items moves the estimate by 0.015, and only 3 of the 14 max-entropy items had a parse failure at all (mean entropy 0.99 for parse-failure items vs. 0.864 for clean ones – a small gap, not a confound). Under the harshest simultaneous cut the interval still excludes 0.5. The underlying distribution is monotone rather than concentrated in one cell: of items at zero entropy 11/14 were transcribed correctly, falling steadily to 0/14 at maximum entropy.

That top cell is worth stating as a result rather than treated only as a robustness threat: **every one of the 14 items where all five transcriptions disagreed was wrong** – precision 1.00 as an abstention rule, recall 0.24 of all 58 errors. On $n = 100$ a zero-false-positive cell of size 14 is suggestive rather than established, but it is the most directly actionable thing in the pilot.

4 Phase 1: Reasoning Entropy – Clean Pipeline, Unresolved Signal

Unlike perception, the reasoning/grading pipeline required no comparable fix. It clusters a single parsed binary digit (error present/absent) rather than free text, so it never had the "wrong span" problem. The pipeline was independently re-verified by recomputing correctness from scratch for all 100 items against the stored results: zero mismatches.

4.1 $K = 5$ baseline

Metric	Value
AUROC (reasoning entropy $\rightarrow$ grading error)	0.618
95% CI	$[0.489, 0.741]$
Grading accuracy	75%
Median entropy, correct group ($n = 75$)	0.5004
Median entropy, incorrect group ($n = 25$)	0.5004

This interval includes 0.5. With 25 items in the minority class, an AUROC of 0.618 is not distinguishable from chance, and earlier drafts of this report describing it as a "real but weak" signal overstated it. Everything downstream in this section was motivated by trying to strengthen a number that was never established in the first place.

The medians are also *exactly* tied. With only 5 samples of a binary outcome, there are just 3 realistic entropy values (unanimous, a 4–1 split, a 3–2 split), which structurally limits how well two groups can be told apart by this statistic.

4.2 $K = 15$ follow-up: is the weak signal a real ceiling, or under-sampling?

To test whether $K = 5$ was simply too coarse to reveal a real effect, grading was resampled to $K = 15$ for the same 100 items (reusing the original 5 samples and drawing 10 more), keeping perception untouched.

Metric	$K = 5$	$K = 15$
AUROC	0.618	0.665
95% CI	$[0.489, 0.741]$	$[0.514, 0.800]$
Grading accuracy	75%	82%
Distinct entropy values reached	5	13
Median entropy, correct group	0.500	0.439
Median entropy, incorrect group	0.500	0.608

The exact median tie *does* break at $K = 15$ (Figure 3), and the $K = 15$ interval does clear 0.5 – though only just. A pre-registered decision rule (AUROC ≥ 0.72 *and* the median tie breaking, to count as "sharper"; 0.55–0.72 as "real but modest"; below 0.55 as "noise") classifies this as **confirmed_modest**.

That classification should be read with the paired comparison in mind, because the apparent $0.618 \rightarrow 0.665$ gain is *not* evidence that more sampling helped: the paired difference is -0.047 with a 95% CI of $[-0.178, +0.090]$, comfortably spanning zero. The tie breaking is a genuine structural consequence of having 13 reachable entropy values instead of 5, but tripling K bought no measurable discrimination. Note also that grading accuracy itself moved (75% $\rightarrow$ 82%), so the two AUROCs are scored against different correctness labels – 25 errors vs. 18 – which the paired interval cannot correct for.

This pointed at a different question: is more sampling the right lever at all, or is the *digit itself* too coarse a readout of the model's judgment?

4.3 Reasoning-text clustering: a promising readout, an unproven improvement (unresolved)

The grading prompt asks the model to write a `**Reasoning:**` explanation before its `**Error:**` digit. Every analysis so far discarded that explanation and kept only the digit. Clustering the reasoning text instead – via the same bidirectional NLI entailment approach that failed for perception (Attempt 2) – was tested against the already-collected $K = 5$ raw samples, at no additional GPU cost.

Metric	Digit-only	Reasoning-text
AUROC	0.618	**0.702**
95% CI	$[0.489, 0.741]$	$[0.579, 0.817]$
Median entropy, correct group ($n = 75$)	0.500	0.950
Median entropy, incorrect group ($n = 25$)	0.500 (tied)	1.332
Paired difference (text − digit, same items, same labels): $+0.083$, 95% CI $[-0.064, +0.233]$		

The improvement is not statistically established. Reasoning-text clustering at $K = 5$ does clear chance on its own terms ($[0.579, 0.817]$ excludes 0.5) where the digit-only baseline does not ($[0.489, 0.741]$ does not) – but that pattern is precisely the "difference in significance is not a significant difference" trap. The correct test is the paired one, and its interval spans zero. An earlier draft of this report called this "a stronger, less noisy signal" and "arguably the most interesting result of the reasoning-arm investigation"; on the evidence available at $n = 100$, that claim is not supported. What survives is narrower and still worth having: reasoning-text clustering is the only reasoning-arm readout whose own interval excludes chance, and it has a documented mechanism (below) for why it should be less noisy than the digit. That makes it the right candidate to test at larger n, not a demonstrated win.

Given that the same NLI approach caused Attempt 2's over-merging failure on perception, this result was not trusted without directly checking for the same bias. The check: do any two samples that *disagree on the parsed digit* (one says error, one says no error) ever get merged into the same reasoning-text cluster? They do – 55 times across the 61 items where samples disagreed on the digit. Inspecting a diverse sample of these merges (8 cases, different items) showed 7 of 8 were not opposed reasoning merged by mistake: *both* texts explicitly argued "there is an error," just citing different specific details, while their own digits disagreed. Quantified across the full dataset: **60/494 samples (12%)** have digit$= 0$ while their own reasoning opens by flagging a mistake ("incorrect," "error," "mistake," ...). The model is self-contradicting within single samples – writing a reasoning that argues for an error, then emitting the opposite digit anyway, plausibly decoding noise on the final token, independent of the fully-formed judgment already written. At $K = 5$, reasoning-text clustering is not an artifact of NLI bias; it is a readout that is not vulnerable to this specific noise source the way the digit alone is (Figure 4).

This 12% self-contradiction rate is a measurement about the model, not about the clustering method, and it is unaffected by everything the confidence intervals overturn – it is a direct count over 494 parsed samples, not an AUROC comparison. It stands as an independent finding regardless of whether reasoning-text entropy ever proves to be the better predictor.

4.4 $K = 15$ GPU confirmation: the $K = 5$ result does not generalize (failed)

A GPU-accelerated confirmation at $K = 15$ (impractical on local CPU – a single item's 105 pairwise NLI comparisons took 35s locally; GPU reduced the full 100-item pass to 58s) was run via a dedicated notebook (`pilot/02_reasoning_text_entropy.ipynb`). It reproduced the $K = 5$ result almost exactly (AUROC 0.7019 vs. the 0.702 found locally), then revealed the $K = 15$ result does not hold up:

Metric	$K = 5$	$K = 15$
AUROC (reasoning-text)	0.702	**0.520**
95% CI	$[0.579, 0.817]$	$[0.378, 0.664]$
Items with a digit split	61/100	87/100
Cross-digit cluster merges	55	**1187**
Paired difference ($K = 5 - K = 15$, same items): $+0.182$, 95% CI $[+0.037, +0.330]$		

The $K = 15$ interval includes 0.5, so the technique is not distinguishable from chance at that K. More usefully, the degradation from $K = 5$ to $K = 15$ is the *only* reasoning-arm comparison in this report whose paired interval excludes zero – this is the one place where the data can actually tell two conditions apart. It carries one caveat that the interval cannot absorb: grading accuracy differs between the two runs (25 errors vs. 18), so the two AUROCs are scored against different label sets. The mechanistic evidence below is what makes the degradation credible rather than the interval alone.

The paired comparison against the digit-only baseline at $K = 15$ is -0.145, 95% CI $[-0.317, +0.037]$: directionally worse, not resolvably so. The cross-digit merge count grew far faster than the $\sim 10\times$ increase in available pairs ($\binom{15}{2} = 105$ vs. $\binom{5}{2} = 10$) would predict – a $\sim 21\times$ increase in merges from a $\sim 10\times$ increase in comparisons. Re-clustering individual items locally to inspect actual cluster membership found the cause: **transitive cascade merging**. The clustering method unions any two samples whose reasoning mutually entails, then merges transitively (union-find) – so a single mistaken pairwise link can silently absorb an entire subgroup into a cluster that does not actually agree. One item's largest cluster contained 11 of 15 samples, mixing digit=1 samples arguing *"incorrectly multiplies by 5 instead of 9/5"* with digit=0 samples arguing *"does not contain any apparent...errors"* – genuinely opposed conclusions, merged together. This is not universal: a second inspected item's largest cluster (7/15 samples) was entirely digit=0 with consistent reasoning, correctly merged. The failure is concentrated in a subset of items, but at $K = 15$ that subset is large enough to collapse the aggregate signal to near-random. More pairwise comparisons mean more chances for a single false-positive entailment call to cascade, and union-find has no mechanism to catch or bound that once it happens.

The mechanism is established: the cascade was found by re-clustering real items and reading actual cluster membership, and it reproduces as a unit test. What the confidence intervals add is a bound on how much of the *aggregate* $0.702 \to 0.520$ drop can be attributed to it – the paired interval $[+0.037, +0.330]$ is wide, so the degradation is real in direction but poorly determined in magnitude. A structural fix (e.g. requiring a minimum fraction of pairwise agreement within a cluster rather than any transitive chain, or a stricter/larger NLI model) would need to precede any further use of this technique at higher K – though at $n = 100$ such a fix could not be evaluated with any precision either, which is why sample size is the recommended next lever rather than the algorithm.

4.5 Manual inspection: why the reasoning arm underperforms

The confidence intervals establish that the reasoning signal is unresolved but say nothing about *why*. To separate the three plausible causes – clustering, prompt ambiguity, and the model simply being confidently wrong – individual failure cases were read directly, selected from the two informative quadrants: low entropy but graded wrong, and high entropy but graded right.

The grading task is close to degenerate. The 100-item sample is **87/13** on the ground-truth `has_error` label. A constant "there is an error" predictor therefore scores **0.87**, while the model's majority vote scores 0.75 at $K = 5$ and 0.82 at $K = 15$. *The model loses to the trivial baseline.* The model also answers "error" in 383 of 495 parsed samples and for 86 of 100 items, and identifies only **1 of the 13 clean answers** correctly. Grading accuracy is therefore not a meaningful measure of task competence here, and an uncertainty metric layered on top of it inherits the problem.

The dominant failure is confident wrongness, not clustering. Of the 7 items with low entropy (< 0.3) that were graded wrong, 6 have a clean ground-truth answer: the model invented an error and all five samples agreed. Entropy cannot flag these – there is no disagreement to measure. Reading the 6 unanimous cases, they are arithmetic and recall failures rather than defensible disagreements: on $2x^2 - 4x + 3 = 0$ the model asserts $b = -9$ and computes a discriminant from it; on the distance between $(1, 7)$ and $(4, 2)$ it states the formula should be $\sqrt{(1 + 4)^2 + (7 - 2)^2}$; in another it declares $2ab = 29-9$ incorrect and then derives the same value the answer gave. Only one of the six resembles prompt ambiguity (objecting to the *method* used to eliminate a variable rather than to the result) – and even there the model's proposed alternative does not work. **Clustering is not implicated in any of them**: the digit-only pipeline clusters an atomic 0/1 token, and these items are unanimous.

Pooling hides a signal that works. Because the model's bias runs one way, entropy's predictive direction reverses between strata (`pilot.plotting.stratified_auroc`):

Configuration	Stratum	n	n_{wrong}	AUROC (95% CI)
Digit, $K = 5$	pooled	100	25	0.618 $[0.489, 0.739]$
	has error (gt=1)	87	13	**0.756** $[0.618, 0.876]$
	clean (gt=0)	13	12	0.125 $[0.000, 0.250]$
Digit, $K = 15$	pooled	100	18	0.665 $[0.512, 0.799]$
	has error (gt=1)	87	6	**0.891** $[0.813, 0.955]$
	clean (gt=0)	13	12	0.000 $[0.000, 0.000]$
Text, $K = 5$	pooled	100	25	0.702 $[0.579, 0.817]$
	has error (gt=1)	87	13	0.770 $[0.635, 0.890]$

Within the majority stratum the digit-only signal is considerably stronger than anything reported elsewhere in this document, and it *improves* with K ($0.756 \to 0.891$) – reversing, within this stratum, the pooled finding that $K = 15$ bought nothing. Within the 13 clean items it is inverted: the model is confidently wrong there, and the single item it graded correctly is the one where it was most uncertain. Pooling two strata whose predictor runs in opposite directions does not produce a weaker version of the real effect; it produces a different quantity that describes neither.

This is a hypothesis, not a result. The stratification was chosen after seeing the pooled numbers, so it is post-hoc subgroup analysis and is not protected by the confidence intervals shown. The $K = 15$ majority stratum rests on only 6 misgraded items, where bootstrap intervals are optimistic, and the 13-item clean stratum is far too small to estimate anything from. The defensible claim is narrower than the table looks: *the reasoning metric's weakness is attributable to the model's systematic error-present bias on a severely imbalanced label, not to the entropy measure or its clustering*, and the stratified comparison should be pre-registered and re-run on the larger sample rather than reported as a finding now.

Consequence for the study design. A grading task where 87% of items share one label cannot support this question well regardless of n. The scale-up should rebalance the grading subset toward roughly 50/50 on `has_error` – otherwise a larger sample buys tighter intervals around a quantity that is still mostly measuring class imbalance.

5 Phase 2: The $n = 300$ Balanced Run

Phase 1 left two questions it could not answer. Four of its five comparisons were unresolvable at $n = 100$, and its most interesting reasoning-arm claim – that entropy works within the has-error stratum and is merely cancelled by pooling – was a subgroup analysis chosen after seeing the data, resting on 13 clean items. Phase 2 was designed to settle both.

5.1 Design

Two changes from Phase 1, plus a pre-registration:

- $n = 100 \rightarrow 300$, taking the 95% CI half-width on a single AUROC from about ± 0.12 to about ± 0.06.
- **87/13 → 50/50 on `has_error`.** A census of FERMAT (2244 items, 340 clean) confirmed a balanced sample of this size was feasible. Balancing does two things at once: it removes the confound of §4.5, and it *increases power*, because the model over-predicts "error" and so clean items are the ones it gets wrong. A 50/50 sample at $n = 290$ buys the same precision as an unbalanced $n = 480$.
- **Thresholds fixed before the run** (`SCALEUP_PREREGISTRATION`, registered 2026-08-02), including the minimum minority-class size of 30 below which no verdict is read at all. This is what converts Phase 1's post-hoc stratified claim into a real test.

5.2 Perception: replicated and tightened (worked)

	$n = 100$	$n = 300$	Verdict
Pooled	0.788 [0.697, 0.869]	**0.835** [0.787, 0.879]	
`has_error`=1 stratum	0.762 [0.662, 0.855]	**0.830** [0.760, 0.893]	**replicated**
Clean stratum	0.940 [0.762, 1.000] *(13 items)*	0.839 [0.772, 0.898]	

The like-for-like comparison is the `has_error`=1 stratum, since the rebalanced sample is 50% clean against the pilot's 13% and clean items are easier to transcribe. It replicates. The pooled rise from 0.788 to 0.835 sits well inside the pilot's interval and should not be read as an improvement.

Two things worth noting. First, the composition effect this stratification was registered to guard against *did not materialize*: the two strata came out nearly identical (0.830 vs. 0.839), so pooling would have been safe after all. The guard was still correct to register in advance – that it proved unnecessary is only knowable afterwards. Second, the pilot's clean-stratum estimate of 0.940 was noise from 13 items, and settles at 0.839 with 150.

Sensitivity holds at the larger n (`auroc_sensitivity`):

Subset	n	AUROC	95% CI
Full	300	0.835	[0.787, 0.879]
Excluding all parse failures	255	0.839	[0.790, 0.886]
Excluding max-entropy items	239	0.794	[0.737, 0.848]
Excluding both	206	**0.796**	[0.736, 0.852]

5.3 Reasoning: the model cannot do the task (failed)

Metric	Value	
Grading accuracy, balanced sample	**0.517**	(baseline 0.500)
on error-containing answers ($n = 150$)	0.947	
on clean answers ($n = 150$)	**0.087**	
Answers "there is an error"	**93%** of items	
Reasoning entropy AUROC, pooled	0.522 [0.462, 0.582]	**no signal**
Items at zero entropy (all 5 samples agree)	126/300 (42%)	
Distinct entropy values occurring	3 in practice	

This is the central result of Phase 2 and it is unambiguous. **With the class imbalance removed, the model grades at chance.** Its Phase 1 accuracy of 75% was not competence; it was a strong prior toward "error" meeting a test set that was 87% errors. Given a balanced set, that prior is exposed: it identifies 94.7% of genuine errors and 8.7% of clean answers, which is close to the behaviour of a constant predictor.

Reasoning entropy therefore has nothing to measure. Its distribution is nearly degenerate – 42% of items have all five samples agreeing, and only three entropy values occur at all – and its AUROC is 0.522 with a CI comfortably containing chance.

The correct reading is not that the entropy metric is poorly implemented. It is that **sampling entropy is a measure of self-consistency, and self-consistency only predicts correctness when the model has competence for the disagreement to track.** A confidently wrong model is stable and wrong; entropy sees the stability and infers reliability. This is a property of the technique, not a bug in this implementation, and it applies to any sampling-based uncertainty method.

5.4 The pre-registered stratified hypothesis was untestable

Phase 1's most promising reasoning claim was that entropy scores 0.756 within the `has_error`=1 stratum and is only cancelled by pooling. Phase 2 registered a threshold of 0.70 for it. The outcome:

Stratum	AUROC (95% CI)	n_{wrong}	Verdict
Pooled	0.522 [0.462, 0.582]	145	**no_signal**
`has_error`=1	0.836 [0.626, 0.946]	**8**	**inconclusive_underpowered**
Clean	0.239 [0.128, 0.374]	137	**inconclusive_underpowered**

The power calculation behind the $n = 300$ choice assumed grading errors would spread across both strata. They did not: the model's bias concentrated almost all of them in the clean stratum, leaving only 8 misgraded items among 150 error-containing ones. That is below the registered minimum of 30, so the honest verdict is *untestable*, not 0.836. Reporting the point estimate here would repeat exactly the mistake the pre-registration existed to prevent.

The clean stratum is inverted as predicted (0.239, with the interval excluding 0.5 by a wide margin), which supports the confident-wrongness account of §4.5. But it has only 13 correct items, so it too falls under the registered minimum. *This verdict changed after the run*: the inversion branch of `classify_scaleup_result` was initially skipping the minority-class guard that every other branch applied – a coding inconsistency, not a registered choice. Applying it uniformly moved this from "confirmed" to underpowered. The threshold was not altered; the fix is recorded here because it was made after seeing the data.

5.5 Replications and integrity

Phase 2 reproduces the pilot's incidental findings, which is reassuring about the pipeline:

Check	$n = 100$	$n = 300$
Digit contradicts its own reasoning	12%	11.0%
Transcription parse-failure rate	4.0%	3.6%
Grading parse-failure rate	1.0%	0.9%
Temperature-0 anchor non-zero	0/100	0/50
Max-entropy abstention rule precision	14/14 = 1.000	**57/61 = 0.934**

The abstention rule is the one place where a Phase 1 number was optimistic: perfect precision on 14 items becomes 93.4% on 61. That is the expected direction for a small-sample extreme, and 93.4% remains a usable operating point – it flags 57 of the 182 transcription errors (31% recall) with 4 false positives.

5.6 The deferral rule: risk-coverage, AURC, and a conformal guarantee

The abstention rule above is one hand-picked operating point. The general version of the same idea is a *risk-coverage curve*: order items by entropy, most confident first, and ask what error rate results from answering only the most confident X% and deferring the rest.

Coverage	Items answered	Items deferred	Risk (error rate answered)
12.7%	38	262	**0.079**
33.3%	100	200	0.260
52.0%	156	144	0.372
100% (answer everything)	300	0	0.607

A single number summarizes the whole curve: the *area under the risk-coverage curve* (AURC), the average risk swept across all achievable coverage levels. Lower is better, and it must always be read against the random-deferral baseline (just the overall error rate) since a high base error rate caps how good any AURC can be regardless of how well entropy ranks errors.

AURC (entropy-ordered deferral)	**0.410**
Baseline AURC (random deferral order)	0.607
Improvement	0.197

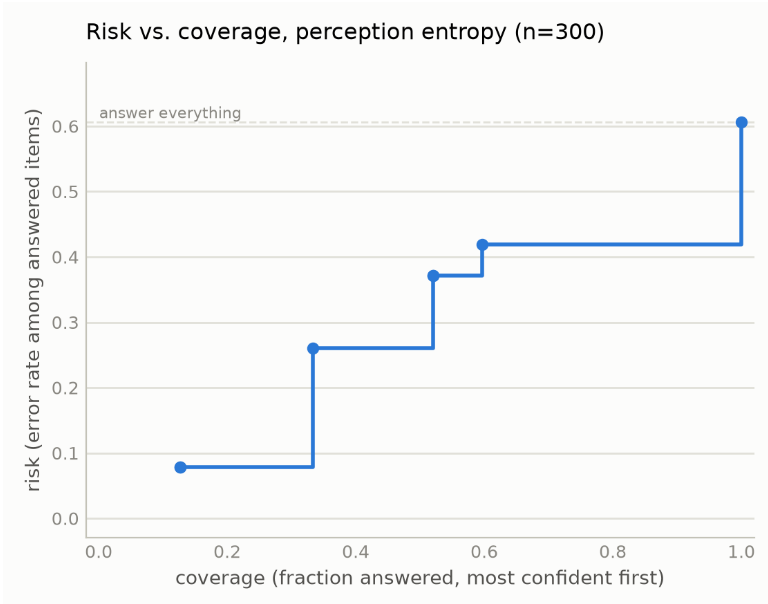

Figure 1: Risk-coverage curve for perception entropy, $n = 300$. A step plot, not a smooth curve, because $K = 5$ entropy has only 7 achievable operating points – the curve is genuinely flat between them and jumps at each one. Coverage 1.0 (answer everything) recovers the overall error rate, 0.607.

Does the guarantee hold on data it has not seen? A hand-picked cutoff carries no guarantee – it is only known to have worked on the data it was chosen from. A *conformal* threshold is calibrated on one half of the items and evaluated on the other half, repeated over 1,000 random splits, to check whether it actually controls risk on held-out data. Targeting at most 30% risk:

Target risk	30%
Mean coverage achieved	12.6%
Mean risk on held-out data	**13.7%**
Splits where the target was violated	2.4%
Splits where no threshold could reach the target at all	**66.7%**

The guarantee holds – observed risk (13.7%) sits well under the 30% target, and it is violated in only 2.4% of splits, both consistent with a conservative, working conformal procedure. The honest caveat is the fourth row: two-thirds of splits could not even attempt the target, because $K = 5$'s seven operating points do not include one fine enough. This is the same resolution limit noted in §6.1's $K = 10$ result, stated here in the specific terms that matter for deployment – more samples per item would not just raise the headline AUROC, it would unlock finer-grained, more reliable deferral thresholds. `pilot.plotting.risk_coverage_curve`, `aurc`, and `evaluate_conformal_abstention` implement all of this and are unit tested against both synthetic cases and this exact $n = 300$ run.

6 Phase 3: Closing the Baseline Gap and Testing Prompt Fixes

Phase 2 left two things unaddressed: perception entropy had never been compared against an independent uncertainty signal (only against a cruder version of itself, a plain count of distinct answers), and the reasoning arm's 93% "there is a mistake" response rate had never been tested against an alternative prompt. Both were run in one session (`pilot/04_confidence_and_prompts.ipynb`), reusing the Phase 2 sample so results join directly onto it.

6.1 Entropy vs. token confidence: entropy wins decisively (resolved)

Capturing per-token log-probabilities requires `output_scores=True`, which forces the model off its fused attention kernel and onto eager attention throughout, including the vision tower – a 14 GiB allocation that a 16 GiB GPU cannot provide regardless of sampling batch size. This experiment therefore needed a ≥24 GiB GPU (run on an A100); the notebook checks available VRAM and skips the capture on smaller hardware rather than failing.

Five additional transcription samples per item were drawn with scores captured, summarized per sample by the mean transition log-probability (overall fluency) and the minimum (the single least confident token):

Signal	AUROC	95% CI
Entropy ($K = 5$)	**0.835**	$[0.787, 0.879]$
−mean log-probability	0.537	$[0.469, 0.605]$
−min log-probability	0.460	$[0.392, 0.527]$
Paired (entropy − mean log-prob.): +0.297, 95% CI [+0.214, +0.380], **resolved**		

This closes the last open gap in the perception result: entropy is not simply recovering information the model's own next-token probabilities already expose. Mean log-probability barely clears chance on its own and loses to entropy by a wide, resolved margin.

The minimum log-probability figure needs a caveat rather than a clean headline. Its CI sits at or below 0.5, which invites the reading "token confidence is actively misleading" – but 81% of items cluster in a narrow band (−min-logprob > 20.5, i.e. probability $\sim e^{-20}$) that plausibly reflects an artifact of batched generation (padding positions in the shorter sequences of a batch) rather than genuine uncertainty about real content. Splitting on that band does not resolve cleanly either: both halves land inside wide, chance-overlapping intervals (0.467 $[0.393, 0.543]$ on the clustered majority, 0.529 $[0.378, 0.682]$ on the rest). The honest conclusion is that −min-logprob is not a usable signal, not confidently *why* it fails – mean log-probability is the trustworthy half of this comparison, and it alone is sufficient to answer the question this section asks. One methodological note for reuse: these token-confidence values come from a *different* draw of 5 samples than the reference $K = 5$ entropy (the original samples predate logprob capture), not the identical generations – a fair comparison, since both are independent draws from the same model and image, but not a same-output rescoring.

6.2 $K = 10$ modestly sharpens perception, as predicted (resolved)

The same generation pass extends transcription to $K = 10$. Scored against the $K = 10$-recomputed correctness label – the fair comparison, since scoring $K = 10$ entropy against the stale $K = 5$ label answers a different and less meaningful question – the improvement predicted from the Phase 2 leave-one-out subsample check materializes:

	$K = 5$	$K = 10$	Paired difference
AUROC	0.761 $[0.706, 0.813]$	**0.806** $[0.756, 0.854]$	+0.045 [+0.005, +0.087], **resolved**
Distinct entropy values	7	34	

A side effect worth flagging: the "ground truth" majority-vote correctness label is itself not fully stable at $K = 5$ – recomputing the majority with 10 samples flips a handful of items. Correctness labels for this pipeline should be read as an estimate that sharpens with K, not a fixed target.

6.3 No grading prompt variant fixes the yes-bias (screened out)

Three alternative grading prompts were screened against the FERMAT baseline on the first 100 items – a screen, not a full measurement, per a pre-registered rule that only a variant clearing the baseline by a wide margin earns a full 300-item run:

Variant	Accuracy	Answers "there is a mistake"	vs. baseline
Baseline (FERMAT)	0.510	94%	–
Restate result first	0.550	38%	below bar
"Half are correct" framing	0.540	83%	below bar
Elicit 0–100 confidence	0.510	96%	below bar

None clears the bar. The *restate* variant is the interesting negative: it collapses the model's default answer from 94% "mistake" to 38% – a large behavioral shift – while accuracy barely moves (0.510 $\rightarrow$ 0.550). The model is not stuck on a biased default that a better prompt frees it from; it simply swaps which near-constant answer it gives. This weighs against "yes-bias is a fixable prompt artifact" and for a genuine capability ceiling on this 3B model, strengthening the case for testing a larger model (§5.3) as the next real lever on the reasoning arm, rather than further prompt engineering.

7 Integrity Checks

Both entropy signals depend on the sampling and parsing pipeline actually working. Two checks validate that before trusting any entropy number.

Check	Result
Temperature-0 anchor, transcription (2 greedy draws/item, must be ≈ 0)	0/100 non-zero
Temperature-0 anchor, grading (2 greedy draws/item, must be ≈ 0)	0/100 non-zero
Transcription parse-failure rate (before marker-regex fix)	13.4% (67/500)
Transcription parse-failure rate (after marker-regex fix)	4.0% (20/500)
Grading parse-failure rate ($K = 5$)	1.0% (5/500)
Grading parse-failure rate ($K = 15$)	0.9% (14/1500)

The temperature-0 anchor passing at 0/100 confirms the sampling/parsing pipeline is behaving deterministically at temperature 0, as it must. The transcription parse-failure rate was itself a real bug (the extraction regex only matched the exact `**Answer:**` markdown the prompt requested, missing common real-world drift such as LaTeX `\textbf{}` bold, plain text, and a full-width colon); widening the pattern recovered most of the missed samples.

8 Summary

8.1 Final conclusions (Phase 2 + 3, $n = 300$)

Arm	Status	AUROC (95% CI)	Basis
Perception	**Works**	**0.835** $[0.787, 0.879]$	Replicated, robust to all cuts
`has_error`=1 stratum	**Works**	0.830 $[0.760, 0.893]$	Like-for-like vs. pilot's 0.762
harshest artifact cut	**Works**	0.796 $[0.736, 0.852]$	$n = 206$, CI excludes chance
vs. token confidence	**Wins**	$+0.297$ $[+0.214, +0.380]$	Paired, resolved – §6.1
$K = 10$ vs. $K = 5$	**Improves**	$+0.045$ $[+0.005, +0.087]$	Paired, resolved – confirms prediction
Reasoning	**Fails**	0.522 $[0.462, 0.582]$	Model grades at chance (0.517)
prompt variants (screen)	**None fix it**	–	3 tried, none clear bar – §6.3

The reasoning result is a statement about the *model*, not the metric. See §5.3. Perception now has both an independent baseline it beats and a confirmed gain from additional sampling; the reasoning arm's failure has been shown not to be a simple, promptable artifact (§6).

8.2 Phase 1 exploratory results (superseded)

Retained because they document how the Phase 2 design was arrived at, not as standing claims. Every reasoning row here is superseded by §5.3.

Arm	Status	AUROC (95% CI)	Notes
Perception (final)	**Working**	0.788 $[0.697, 0.869]$	Robust to all artifact cuts
Perception (attempt 1)	**Failed**	–	83% pinned at max entropy
Perception (attempt 2)	**Failed**	–	Over-merged, 95% implausible
Reasoning, digit ($K = 5$)	**Unresolved**	0.618 $[0.489, 0.741]$	CI includes chance
Reasoning, digit ($K = 15$)	**Unresolved**	0.665 $[0.514, 0.800]$	No paired gain over $K = 5$
Reasoning, text ($K = 5$)	**Unresolved**	0.702 $[0.579, 0.817]$	Beats chance, not the baseline
Reasoning, text ($K = 15$)	**Failed**	0.520 $[0.378, 0.664]$	Cascade over-merging

Reading that table by point estimate alone reverses two of its conclusions, so the paired comparisons are collected here explicitly. Exactly one of the five differences Phase 1 examined can be resolved at $n = 100$ – and it is a degradation, not a gain:

Comparison	Paired difference (95% CI)	Resolved?
Reasoning text − digit, $K = 5$	$+0.083$ $[-0.064, +0.233]$	No
Reasoning text − digit, $K = 15$	-0.145 $[-0.317, +0.037]$	No
Reasoning digit, $K = 5 - K = 15$	-0.047 $[-0.178, +0.090]$	No
Reasoning text, $K = 5 - K = 15$	$+0.182$ $[+0.037, +0.330]$	**Yes**
Perception − best reasoning arm	$+0.086$ $[-0.073, +0.243]$	No†

† Perception and reasoning predict errors on *different tasks* with different base rates, so this row is descriptive only – a paired interval is not the right instrument for it. Perception's standing rests on its own CI excluding chance and surviving §3.4's cuts, not on beating the reasoning arm.

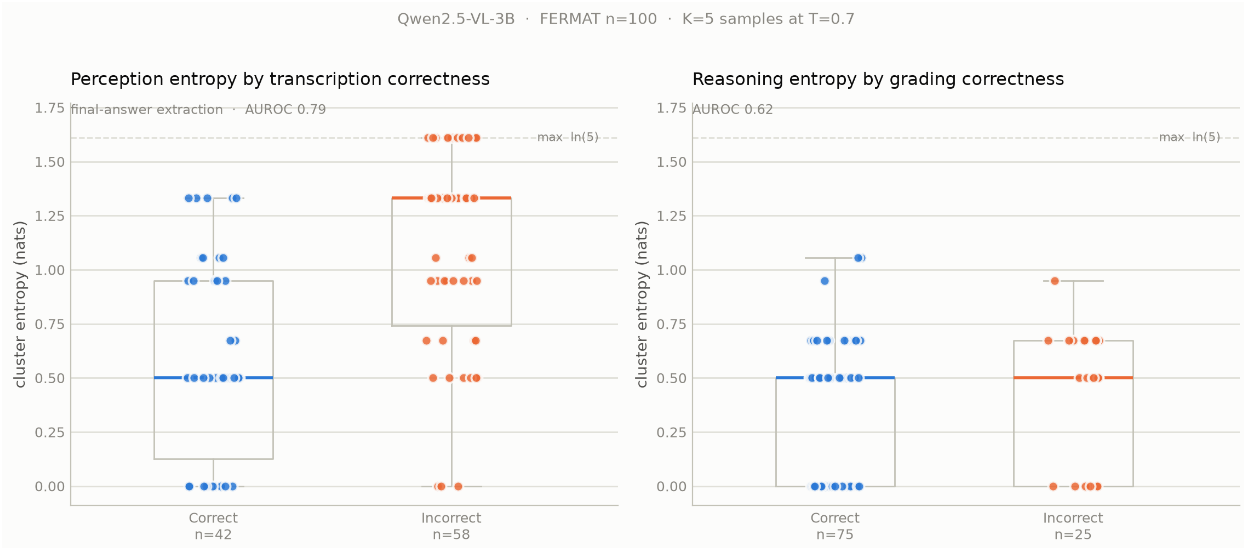

Figure 2: Final result for both arms, $n = 100$. Left: perception entropy by transcription correctness (AUROC 0.788, 95% CI $[0.697, 0.869]$). Right: reasoning entropy by grading correctness at $K = 5$ (AUROC 0.618, 95% CI $[0.489, 0.741]$ – includes chance).

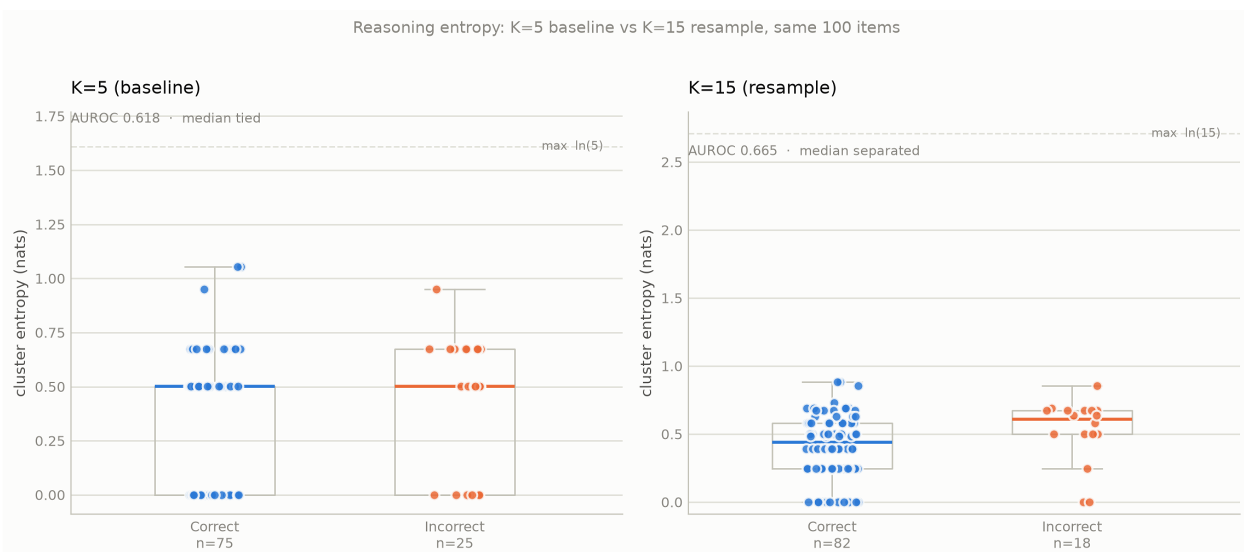

Figure 3: Reasoning entropy before ($K = 5$) and after ($K = 15$) resampling, same 100 items. The exact median tie at $K = 5$ breaks at $K = 15$ – a structural consequence of 13 reachable entropy values instead of 5 – but the paired AUROC difference is -0.047, 95% CI $[-0.178, +0.090]$: no measurable gain in discrimination.

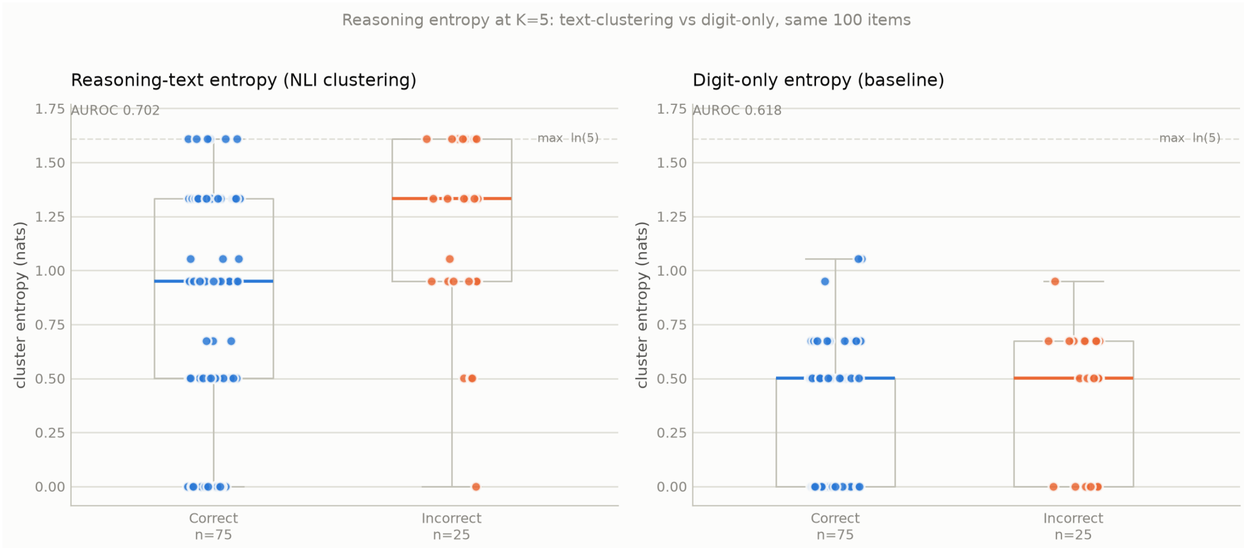

Figure 4: Reasoning entropy at $K = 5$: clustering the model's reasoning text (left, AUROC 0.702) vs. the digit-only baseline (right, AUROC 0.618), same 100 items. The reasoning-text approach separates the incorrect group's median above the correct group's; the digit-only approach ties them exactly. The visible gap is not statistically resolved, however: the paired difference is +0.083, 95% CI $[-0.064, +0.233]$ (§4.3).

9 Limitations

- **The reasoning result is about this model, not about entropy in general.** Qwen2.5-VL-3B cannot grade handwritten math above chance, so this study establishes that entropy fails *in that regime*. It is not evidence that reasoning entropy would fail for a model that can actually do the task. That is the single most important caveat on the negative result, and it is exactly what a larger model would test.

- **Single model.** All results are Qwen2.5-VL-3B; no comparison to a larger model (7B). Given the above, this limitation now carries more weight than it did in Phase 1. (The baseline-uncertainty-method gap mentioned in earlier drafts of this limitation is resolved: entropy now has a measured, resolved win over token confidence, §6.1.)

- **The stratified reasoning hypothesis remains untested, not refuted.** The model's bias left 8 misgraded items among 150 error-containing ones – under the registered minimum. Nothing in Phase 2 licenses a conclusion about it either way.

- **Phase 1 numbers are exploratory.** With a minority class of 18–58 items, its CIs span roughly ±0.12, and four of five comparisons were unresolvable. Several of its analyses were also chosen after seeing data. They are retained to document the path to the Phase 2 design, not as standing claims.

- **Confidence intervals were added retrospectively.** The experiments were designed, run, and initially written up without them; the intervals were computed afterwards on the same data. They correct the report's claims but cannot retroactively supply the statistical power the experiments were never sized for.

- **Perception correctness label is strict.** Transcription correctness requires the extracted final answer to match ground truth after canonicalization; some "incorrect" classifications are plausibly extraction-span mismatches (the model's answer and the ground truth landing in different canonicalization tiers) rather than genuine disagreement, which would mean the true accuracy is somewhat higher than 42%. This biases the reported accuracy conservatively, not favorably.

- **No human double-grading.** Ground-truth labels are taken as given from the dataset.

- **One post-hoc change was made to the Phase 2 analysis code.** The minority-class guard was applied to the inversion verdict after the run, correcting a branch that had been skipping it. The registered threshold was not altered, but the change was made with the data visible and moved one verdict from "confirmed" to underpowered.

- **Reasoning entropy's ceiling is model- and task-specific, and three alternative prompt designs were screened and rejected.** Restating the result before judging, an explicit balanced-base-rate framing, and eliciting a 0–100 confidence were all tested at $n = 100$ (§6.3); none cleared the pre-registered screening bar. This does not rule out every possible prompt design, but it does rule out the simplest fixes, and the *restate* variant's behavior (a large response-distribution shift with no accuracy gain) argues for a genuine capability ceiling over a promptable artifact.

- **Reasoning-text clustering has a confirmed mechanical defect at $K = 15$.** Tested and found to collapse to near-random (AUROC 0.520) due to transitive cascade over-merging (§4.4). The mechanism is verified by direct inspection of cluster membership; the magnitude of its effect on aggregate AUROC is not well determined at this n. The underlying cross-encoder is also a small, general-purpose NLI model rather than one tuned for mathematical reasoning text, which may independently limit how far this approach can go even after the cascade issue is fixed.

10 Recommended Next Steps

Sample size is no longer the binding constraint – Phase 2 resolved what it was built to resolve. Phase 3 closed two of the remaining gaps: perception now has an independent baseline it beats, and the simplest prompt fixes for the reasoning arm have been tried and rejected. What limits the reasoning conclusion now is *model capability*, and there is exactly one experiment left that can resolve it.

1. **Re-run the reasoning arm on a model that can actually grade.** This is the last open item and it directly addresses the main caveat on the negative result. The prerequisite is a capability check, not an entropy measurement: run the grading prompt on the balanced 300 items with a stronger model (Qwen2.5-VL-7B, or a text-only model given the ground-truth transcription) and check accuracy against the 0.500 baseline *first*. If it still grades at chance, reasoning entropy is unanswerable on this task and the negative result stands as general for this task family – now a stronger claim than it was after Phase 2 alone, since §6.3 has already ruled out the task being trivially fixable. If it clears roughly 0.65–0.70, the entropy question becomes live again and the existing analysis code applies unchanged. Blocked on access to compute large enough to run a 7B model, not on anything methodological.

2. **Perception is a deployable result, and the deferral mechanism is now written up alongside the AUROC** (§5.6). It has AUROC 0.835, survives every artifact and scoring-strictness cut, holds across handwriting quality and image quality, beats an independent baseline by a resolved margin, and improves further with K. The remaining open item on this side is narrower than a full write-up: about two-thirds of conformal calibration splits could not reach a 30% risk target at all, because $K = 5$ gives too few operating points. $K = 10$ (§6.1) is the direct, already-tested fix; drawing it for the full $n = 300$ sample rather than the smaller confidence-baseline subset would let the deferral analysis use it too.

3. **The pairwise NLI matrix is already saved** for the $n = 300$ run, so any clustering-variant experiment – including a fix for the cascade fragility of §4.4 – is now a free offline computation. Still not worth doing before step 1: there is no reasoning signal for a better clustering to recover unless a capable model produces one.